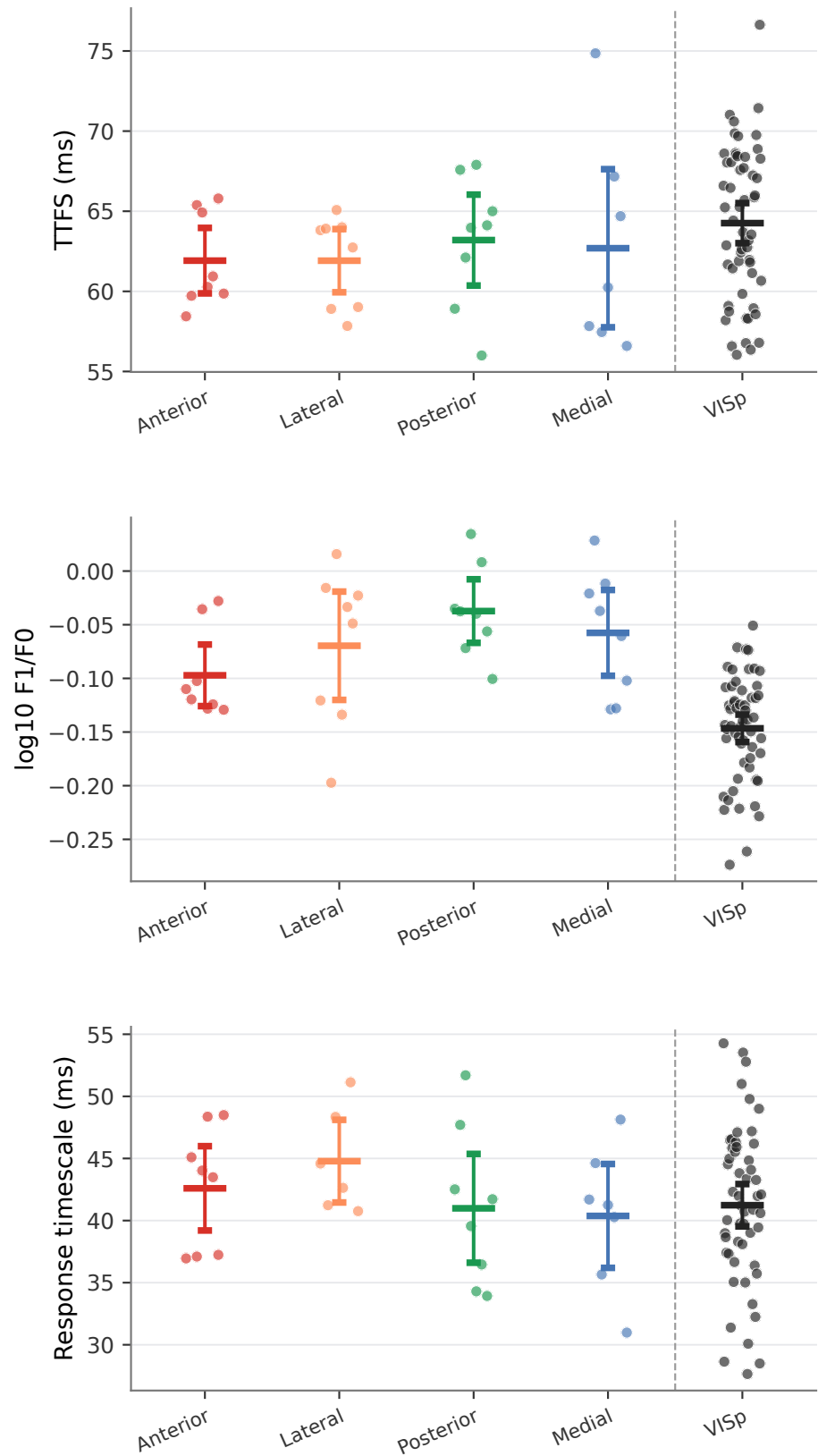


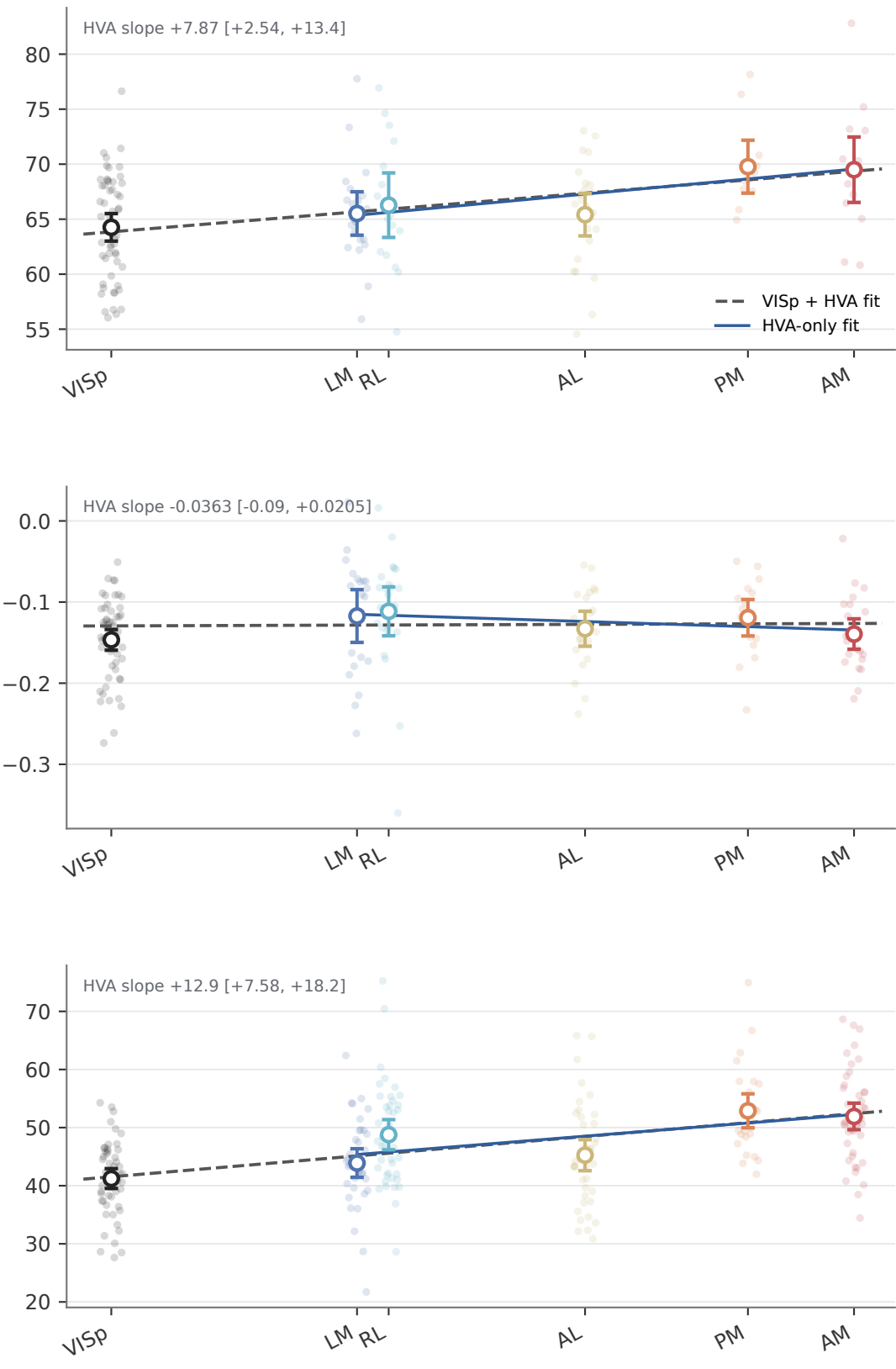
Variant A — hierarchy-led hybrid

Raw session means and hierarchy-score profiles lead; the direct HVA–V1 test remains visible.

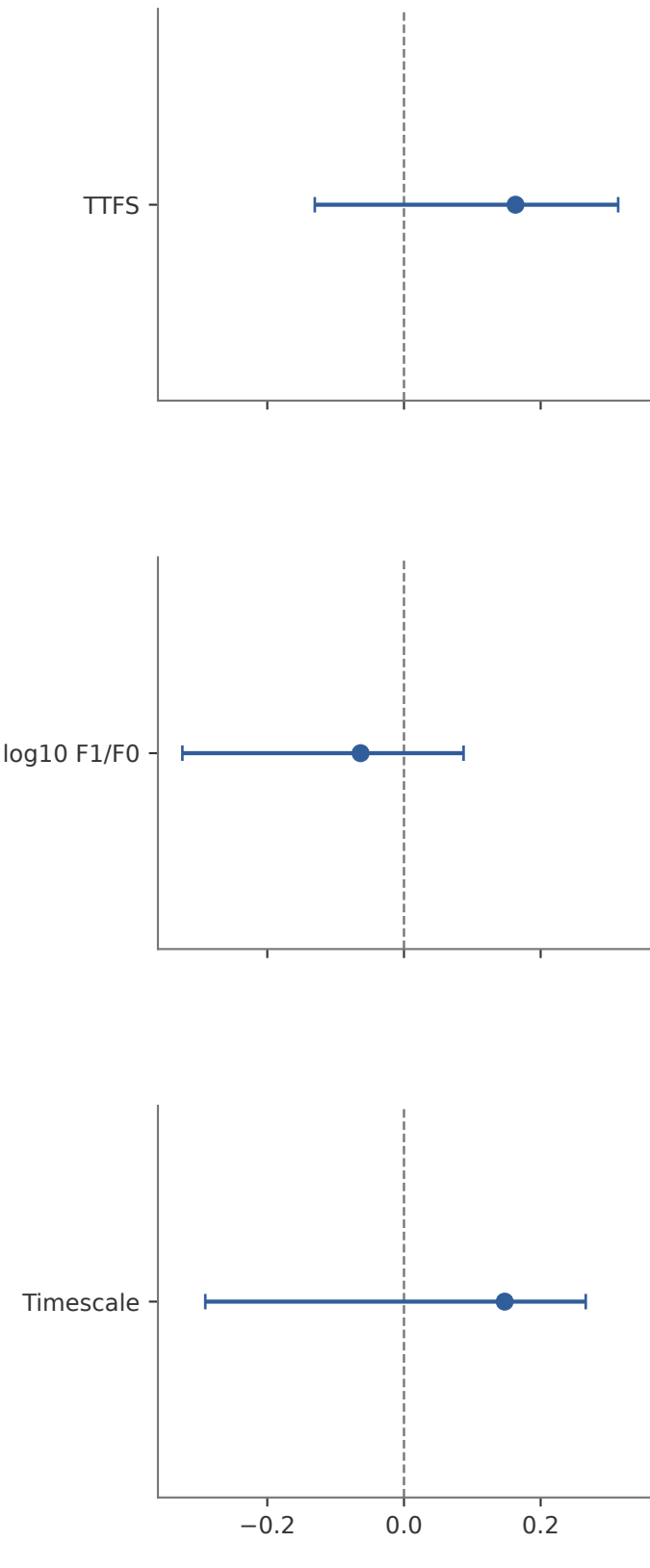
A V1 locations and external VISp reference



B Visual areas by published hierarchy score



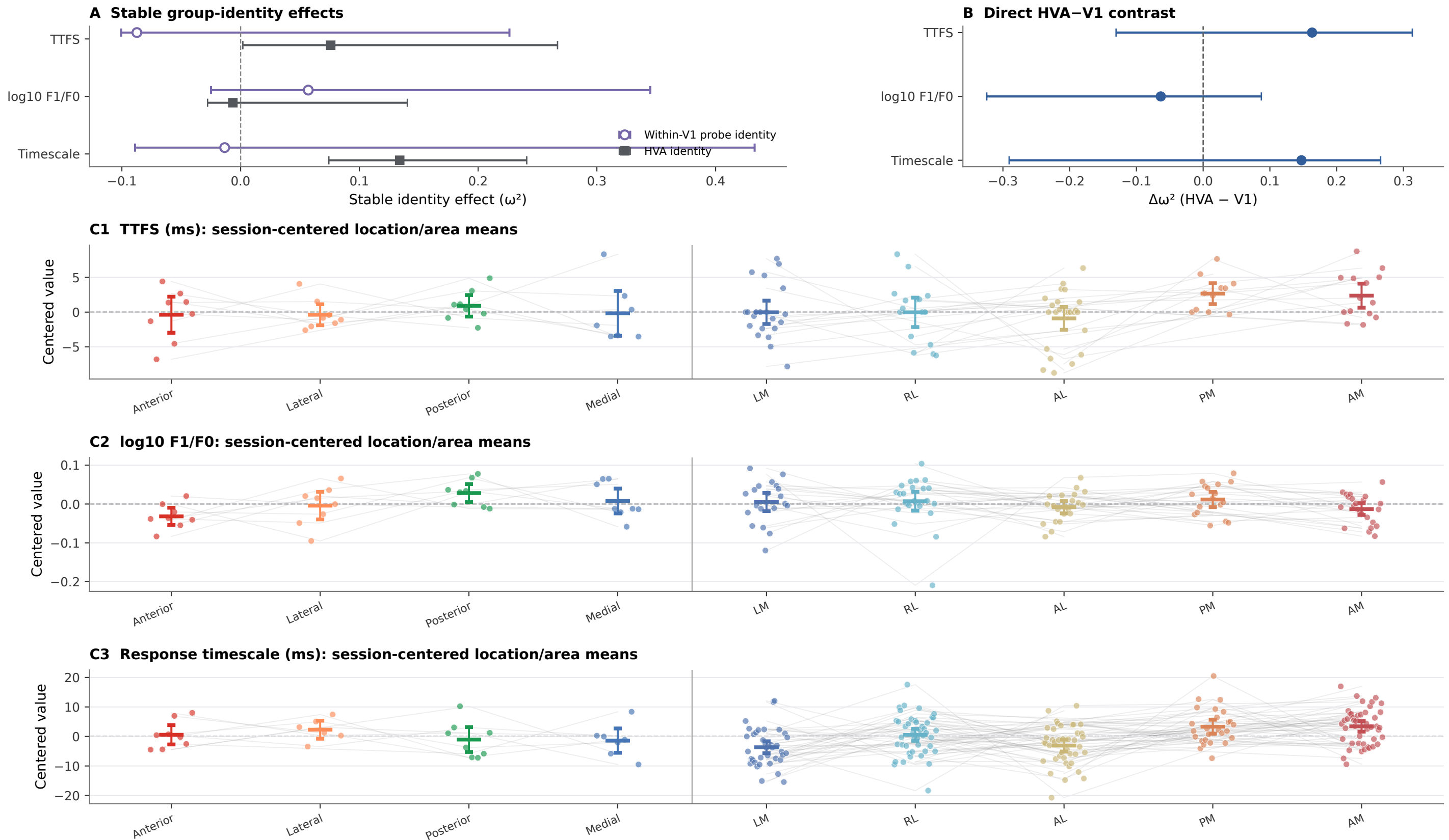
C Direct identity contrast



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant B — benchmark-led

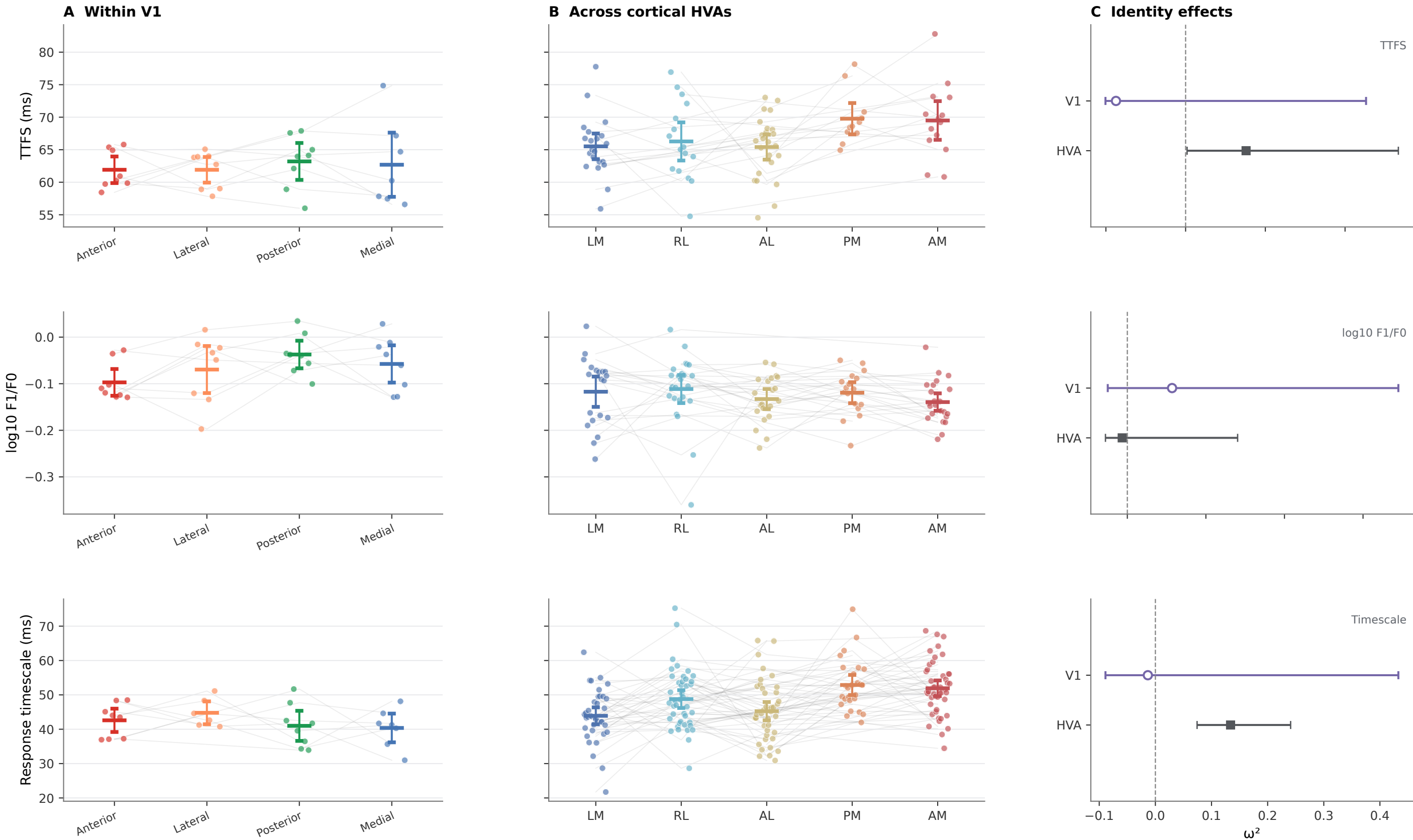
The inferential comparison leads; session-centered data show the observations behind it.



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant C — matched raw-data comparison

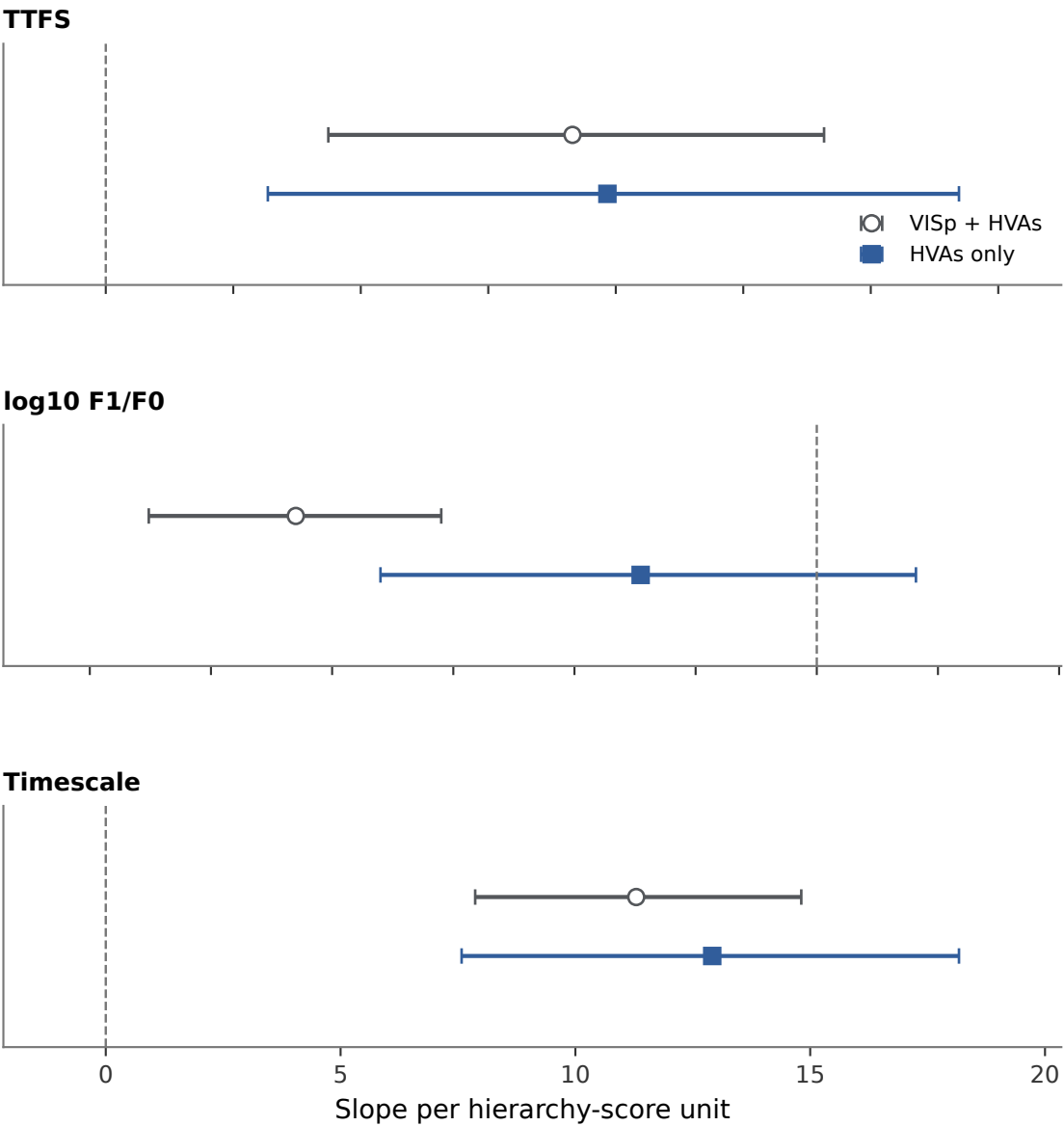
Within-V1 and across-HVA session means share row-wise scales; identity effects provide the summary.



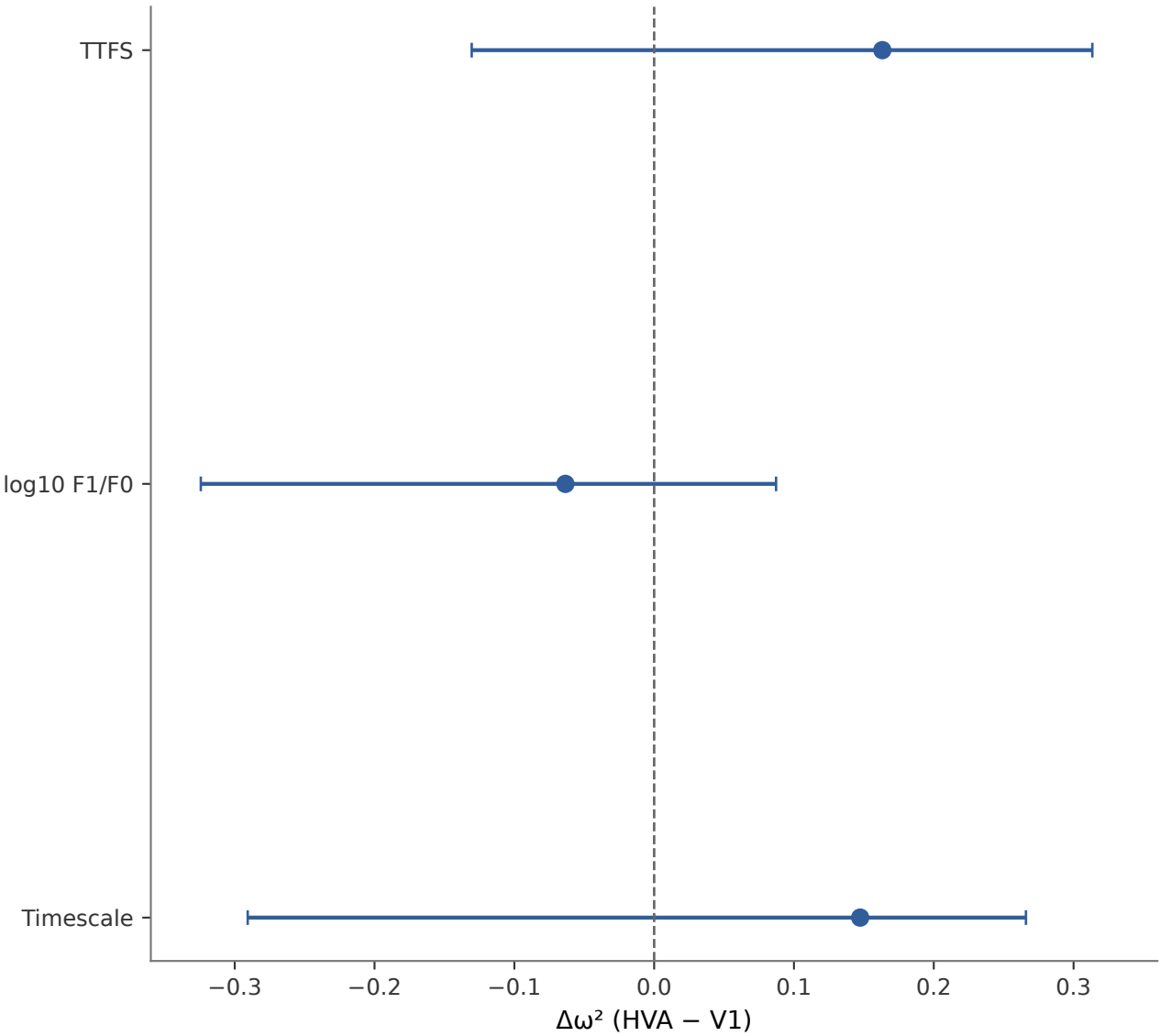
Variant D — compact manuscript summary

Two uncertainty panels for a small main-text footprint; raw session-level panels would move to Extended Data.

A Hierarchy-score associations



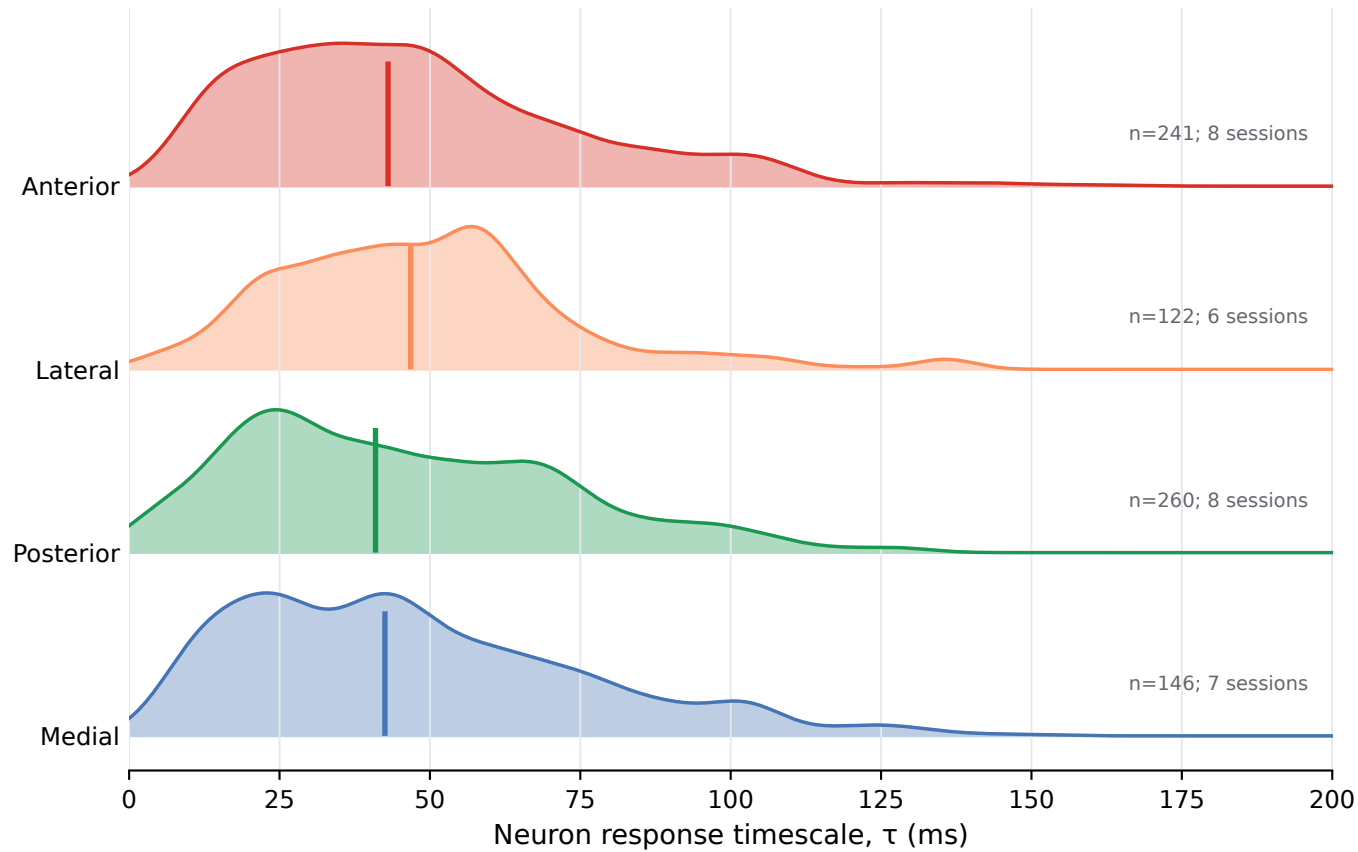
B Does HVA identity exceed within-V1 identity?



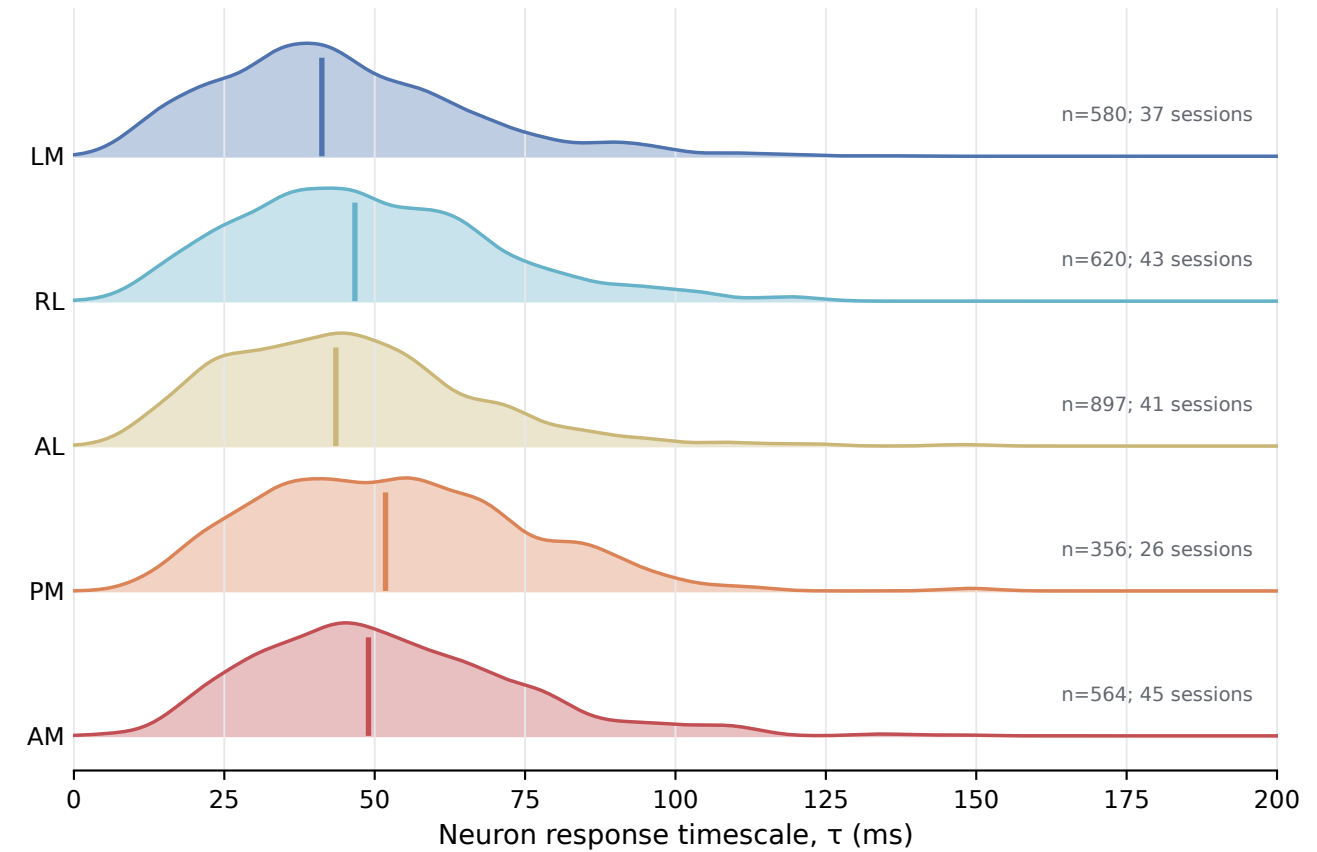
Variant E — matched timescale distributions

Single-neuron ridgelines retain equal session/neuron weight; lower panels show session-level replication.

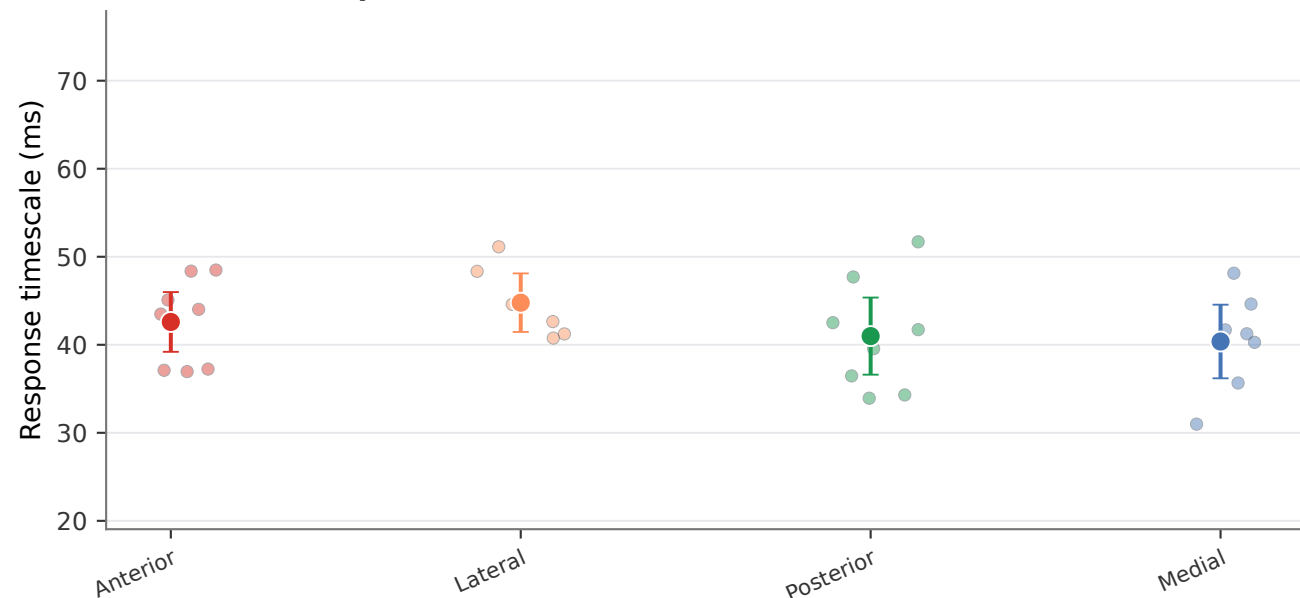
A Full neuronal distributions within V1



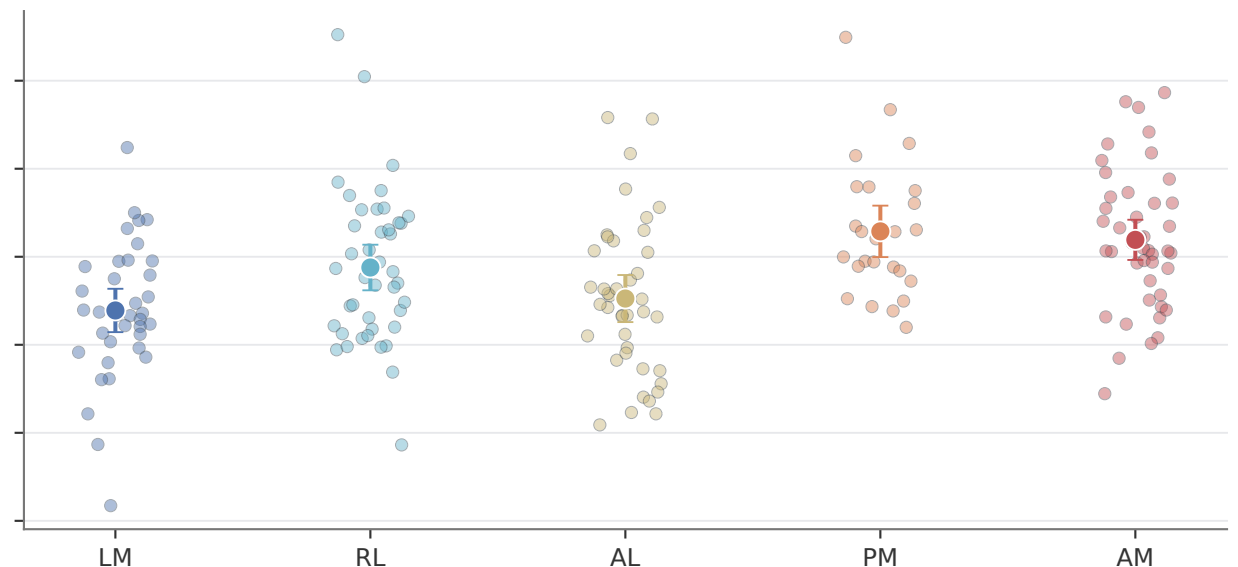
B Full neuronal distributions across HVAs



C V1 session × probe means



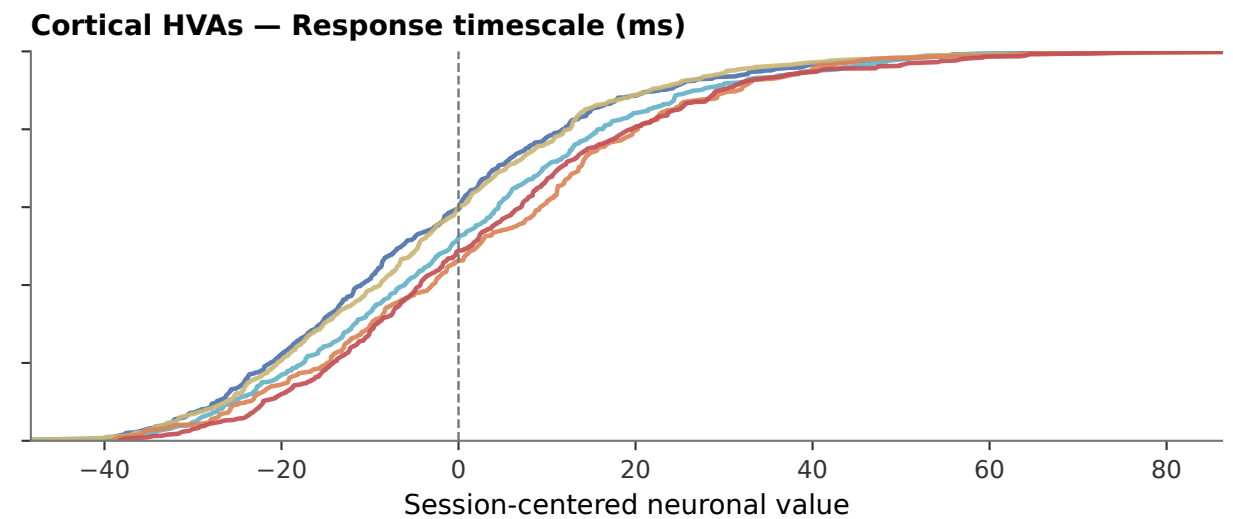
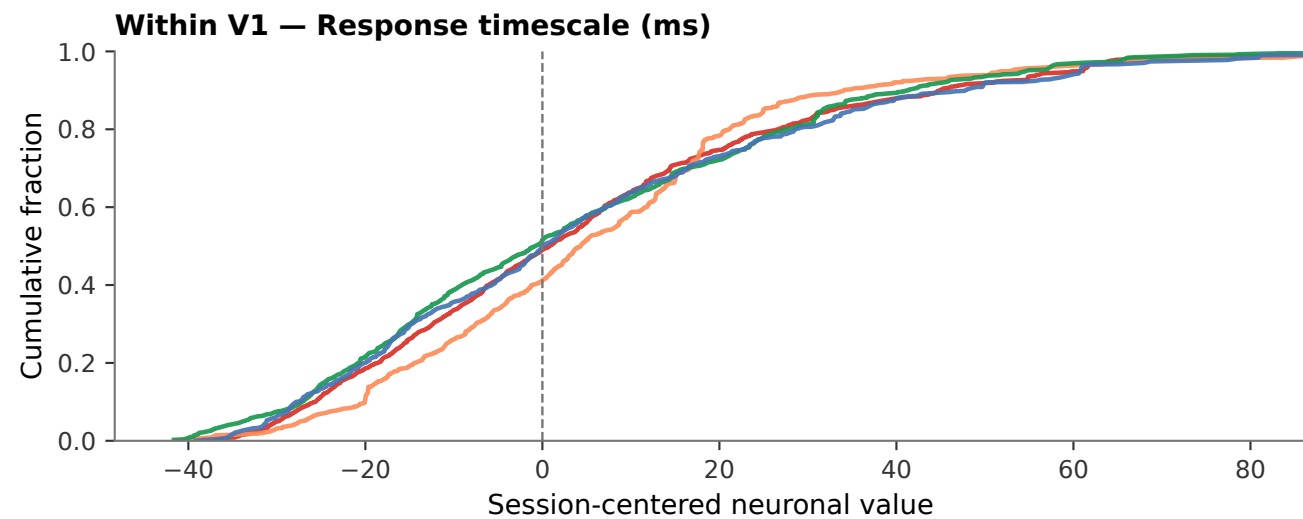
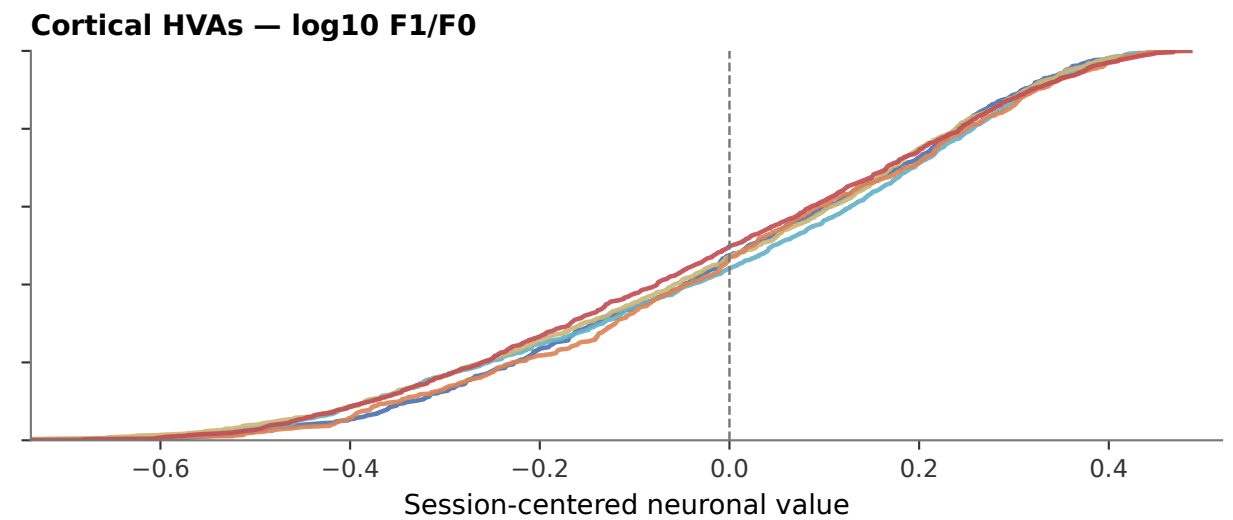
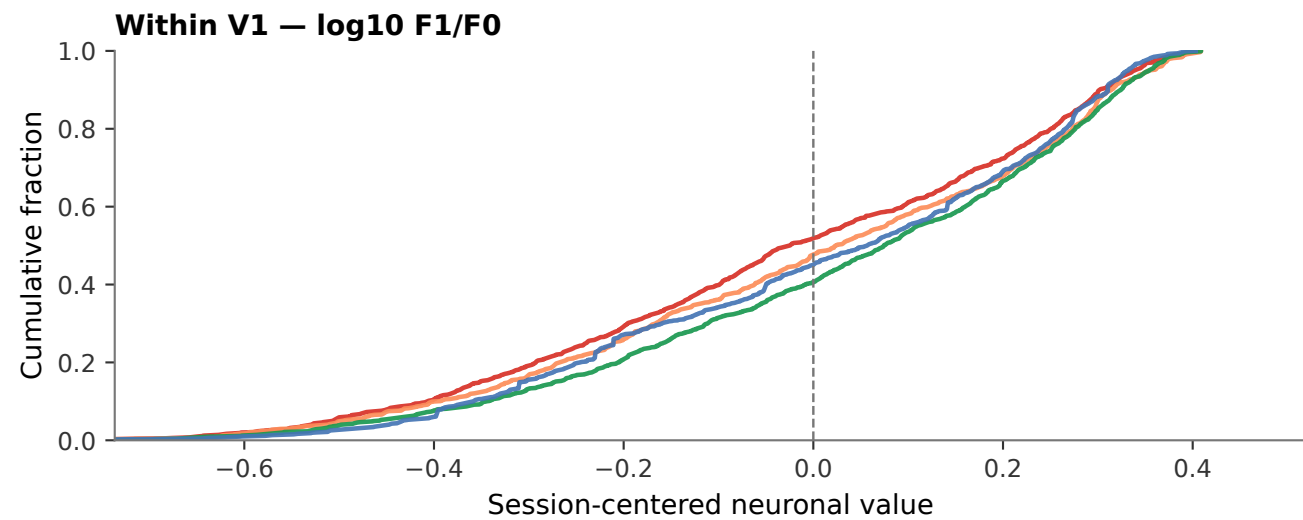
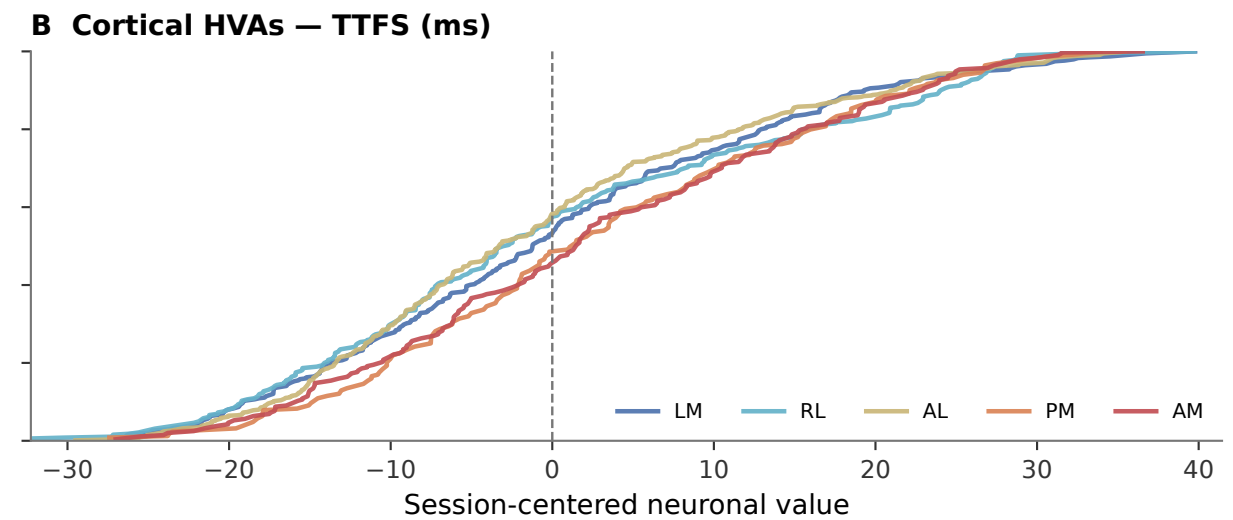
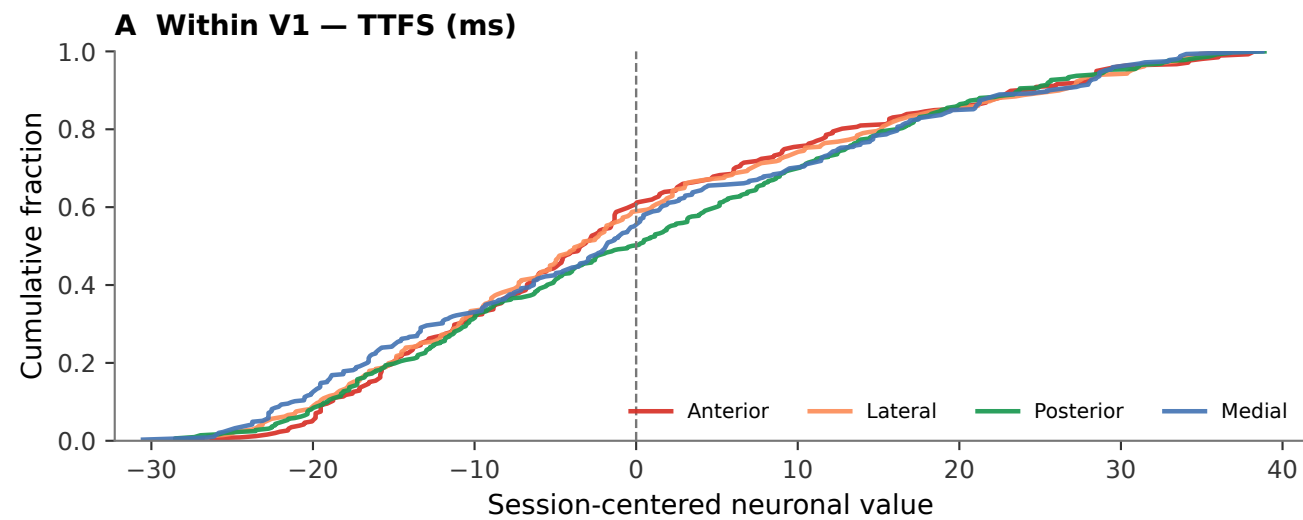
D HVA session × area means



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant F — three-metric full-cell ECDFs

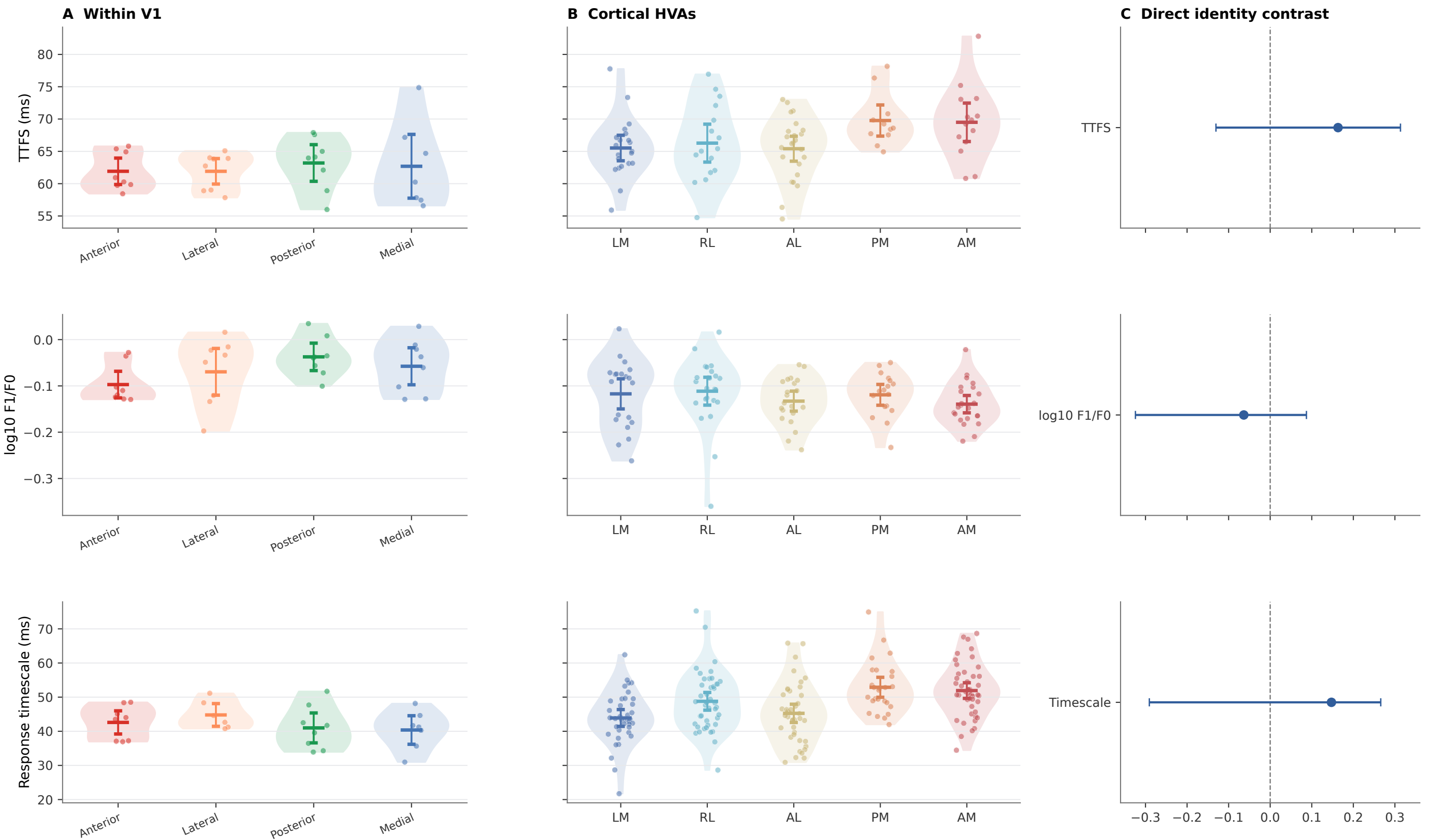
Cells are centered by the frozen session mean and weighted so every session and neuron contributes equally.



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

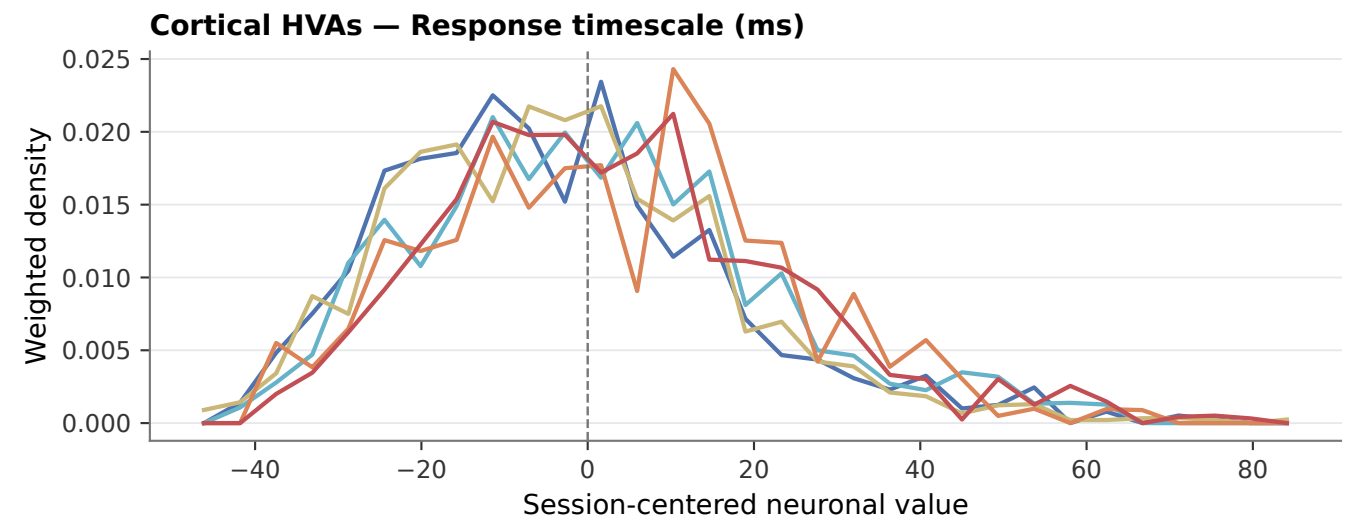
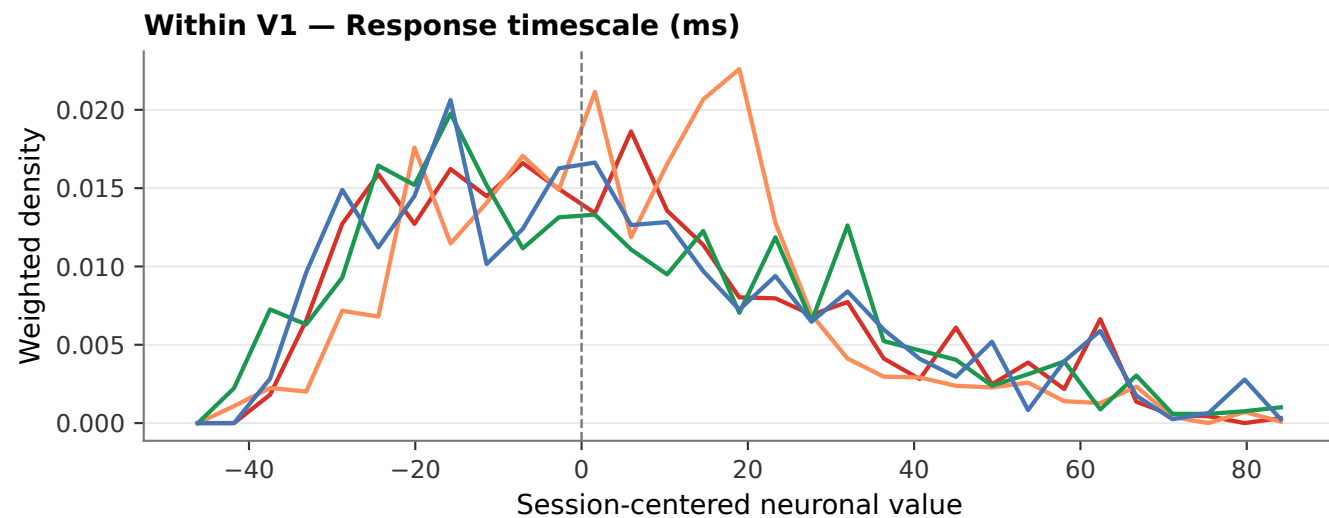
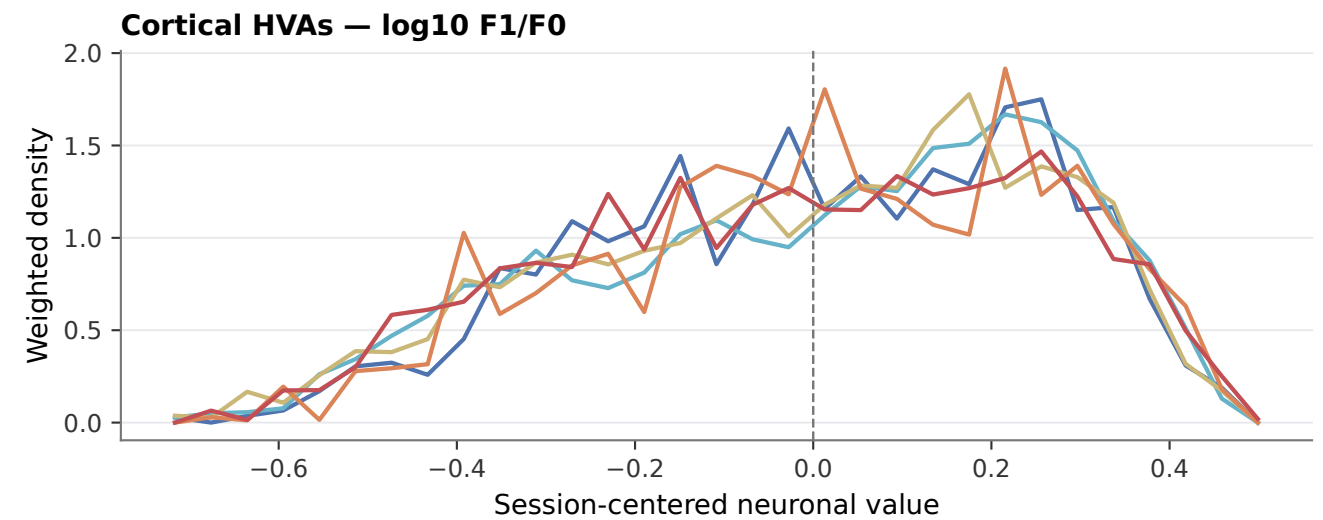
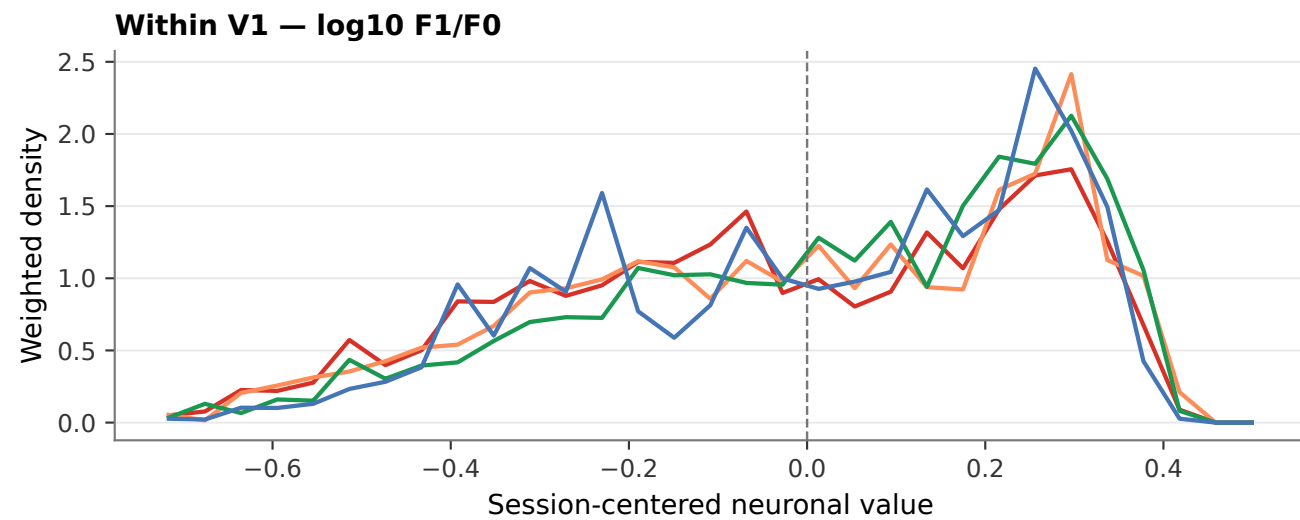
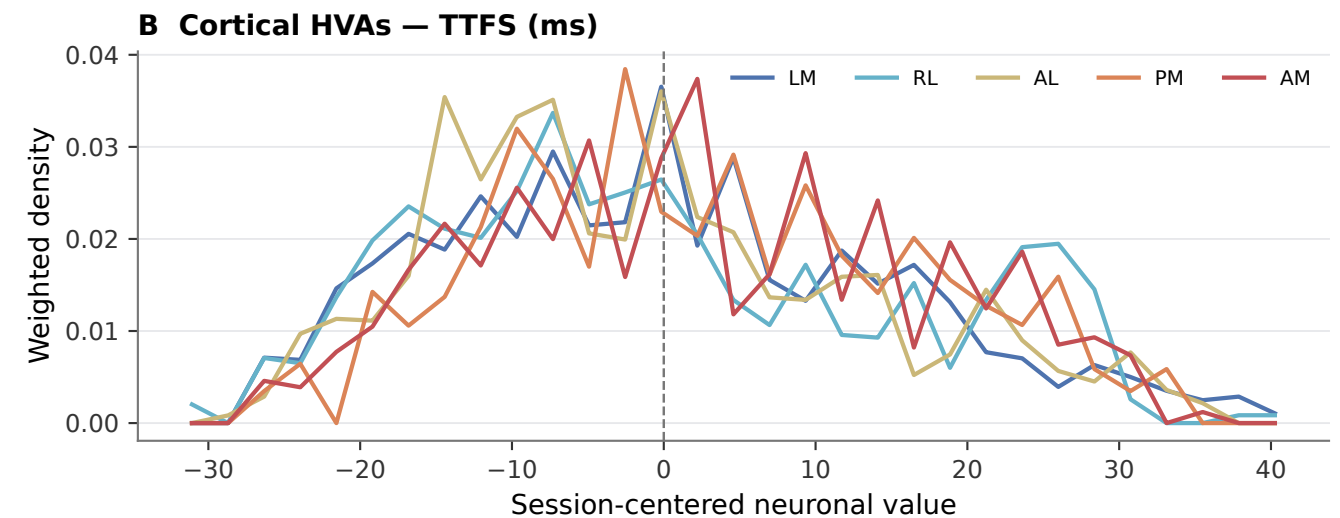
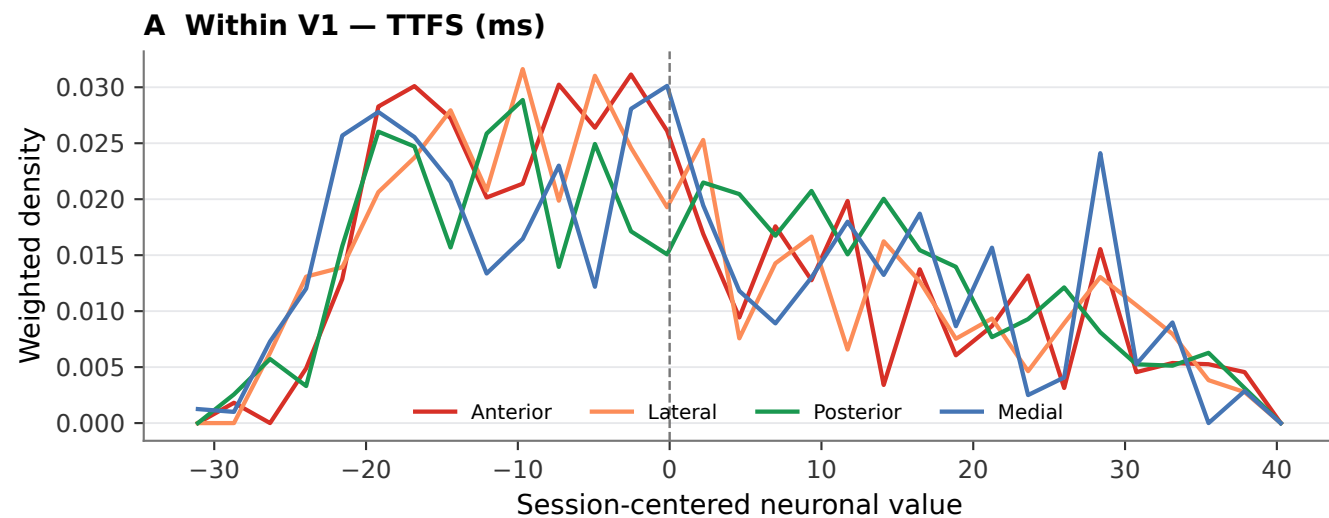
Variant G — session-level violin comparison

Violin envelopes are descriptive; dots are the inferential session × group observations.



Variant H — single-cell histogram small multiples

Explicit bins show distribution shape; weights divide repeated draws and equalize neurons and sessions.



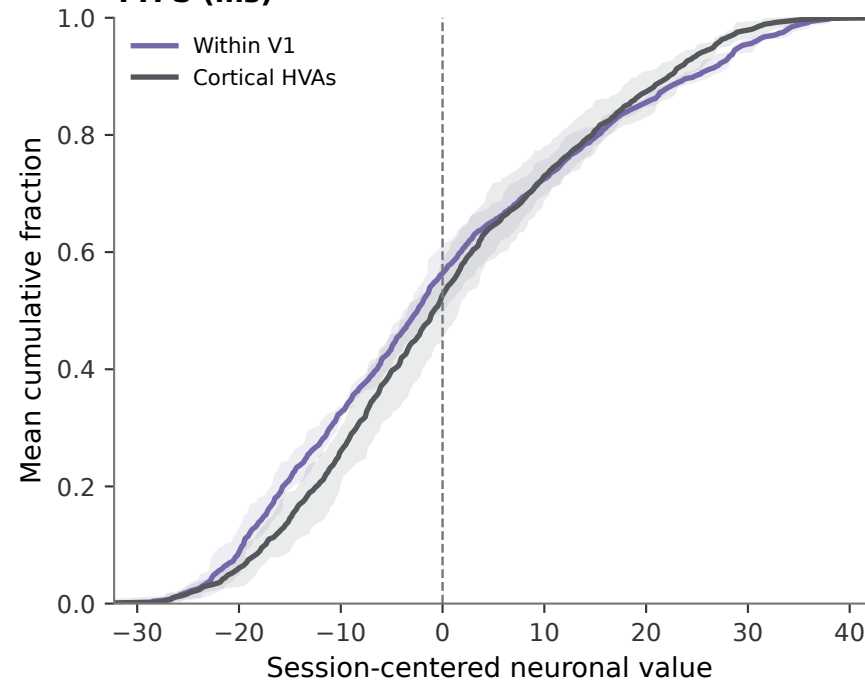
Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant I — compact distribution evidence stack

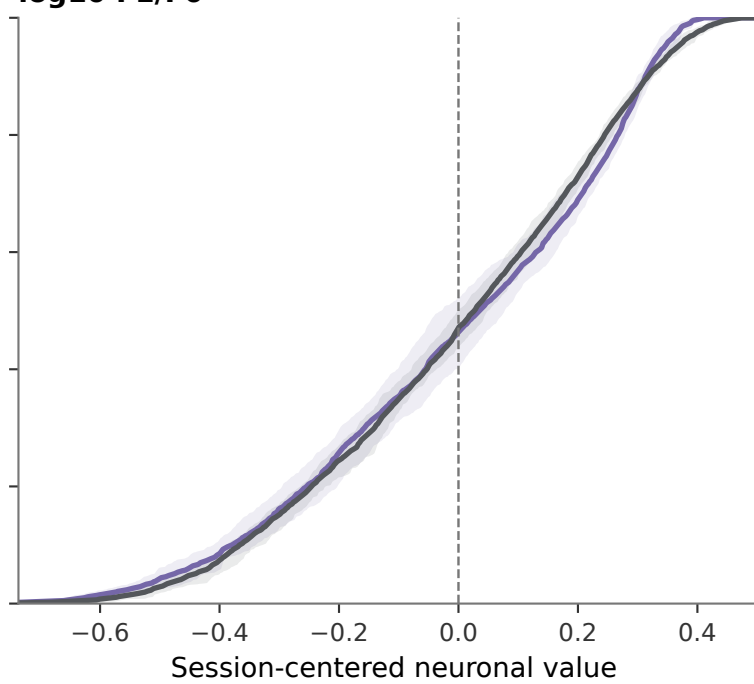
Top: group-balanced neuronal ECDF envelopes. Bottom: every session × location/area mean.

A Full-cell distribution envelopes

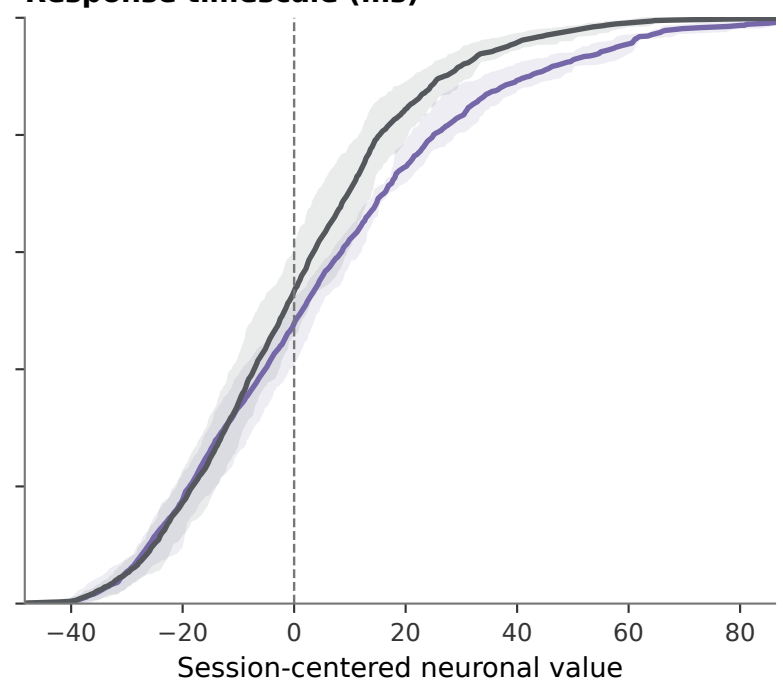
TTFS (ms)



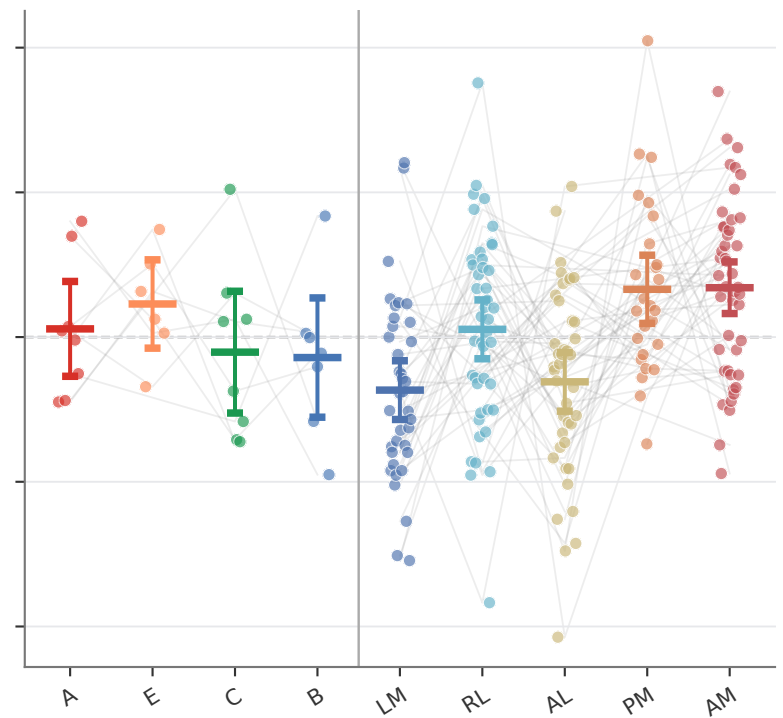
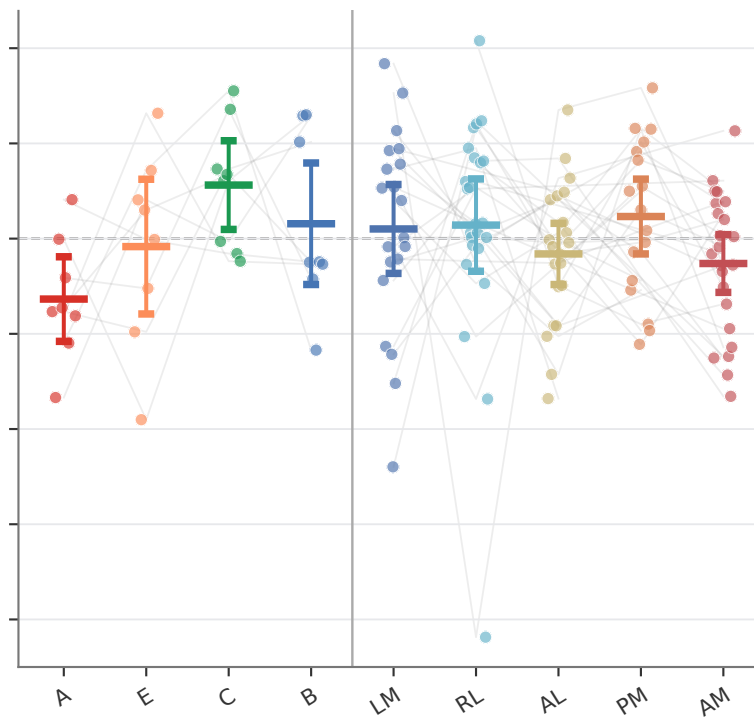
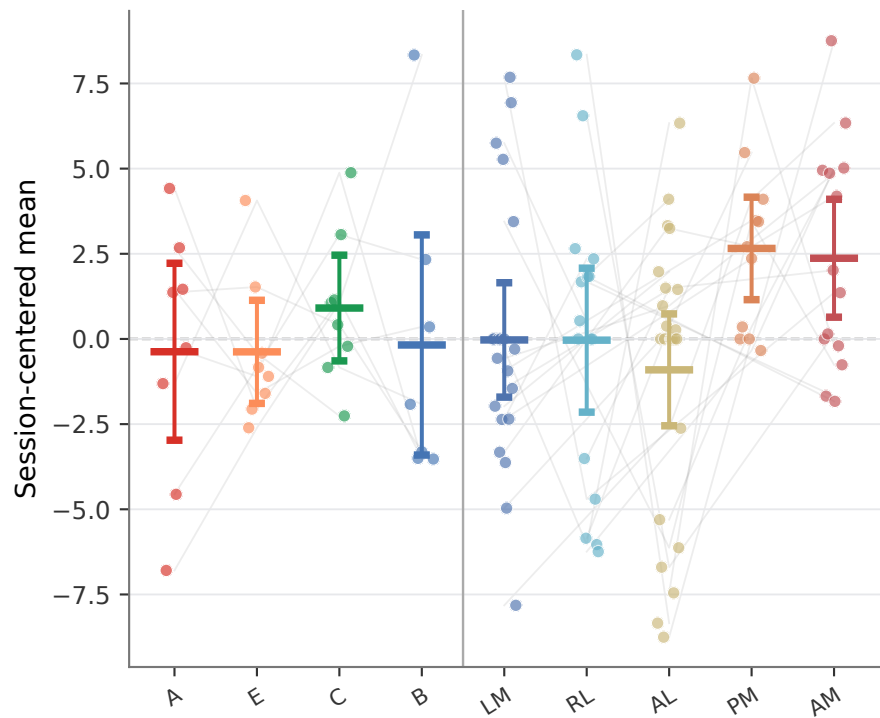
log10 F1/F0



Response timescale (ms)



B Session-level replication



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

